

Local Resolution and Abstention:

A Controlled Comparison of Anisotropic Resolution Fields with Local-Metric and Prototype Baselines under a Predefined Criterion

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Abstract

Research line HQF001 asked whether an *anisotropic resolution field*, a positive definite matrix $A(x)$ attached to every point of the input space and estimated from the local geometry of the data, carries information about uncertainty that local metrics and prototype methods do not already carry, and whether that information improves abstention decisions. The public record of the line reported that the available comparisons did not establish such a residual gain. This note runs the missing controlled comparison. We define the field, three uncertainty scores derived from it (a Mahalanobis margin to local prototypes under $A(x)$, the log-volume of the resolution ellipsoid, and its anisotropy), and evaluate them as selective classifiers by risk–coverage curves and the area under them (AURC), against k -nearest neighbours, nearest class mean, LDA, QDA, logistic regression, random forests and discriminant adaptive nearest neighbours, each tuned by inner cross-validation, on 8 datasets (four real, four synthetic with controlled anisotropic structure) with 15 paired folds and bootstrap confidence intervals. The criterion for a “residual gain”, fixed before the run, was a lower AURC than the best reference on a majority of datasets with a paired 95% interval excluding zero. It was not met: the anisotropic field was significantly better than the best reference on 0 datasets, significantly worse on 6 and inconclusive on 2. Anisotropy adds information relative to the field’s own isotropic ablation on 3 datasets and removes it on 2; on the class-dependent-covariance problem the field beats k -NN, DANN and random forests, but the same gain is obtained by its Euclidean ablation (local prototypes with a margin score) and QDA is better still. An exact statement isolates what anisotropy could add: with the metric equal to the shared within-class inverse covariance, the resolution margin is a strictly increasing function of the LDA posterior margin for two equiprobable classes, is equal to twice the log-ratio of the two largest posteriors for any number of classes, and its departure from LDA under an arbitrary field is $2(x - \bar{m})^\top (A(x) - \Sigma^{-1}) \delta$, so that only the component of the metric along the prototype difference δ can change the score. The formulation is computational throughout; nothing in it refers to physical, let alone quantum, phenomena. With the predefined criterion unmet, the line is closed as a negative result with numbers.

1 Introduction and scope

A classifier that may abstain trades coverage for accuracy [2, 7, 9]. The trade is governed by a scalar confidence score: examples are accepted in decreasing order of the score and the selective risk is tracked as a function of coverage; the area under this risk–coverage curve (AURC) summarises the quality of the score for a fixed classifier [10, 11]. The simplest scores are the maximum posterior probability of a probabilistic classifier [15], the vote fraction of a k -nearest-neighbour rule [3] and the distance margin of a prototype classifier [17, 22]. Locally adaptive metrics [13, 24] change the geometry in which those distances are measured.

Research line HQF001 proposed to represent uncertainty by an *anisotropic resolution field*: a field of positive definite matrices $A(x)$, one per input point, describing in which directions and at what scale the data around x resolve distinctions between classes, together with an abstention rule derived from it. Its public record stated the objective (“test whether the representation adds information beyond local metrics and prototype-based methods”), the finding (“available comparisons do not establish a residual gain over those references”) and the scope (“the formulation is computational and makes no claim about quantum physical phenomena”). No manuscript existed; the present note reconstructs a precise version of the idea from that record, states one exact result about it, and runs the controlled comparison that the record said was missing, with the success criterion fixed in advance.

Scope. Everything below is a statement about algorithms evaluated on public and synthetic datasets. The word “resolution” names a matrix field; it carries no physical meaning here, and we restate the line’s own disclaimer: nothing in this note refers to quantum phenomena. Every number in the text is produced by the scripts of Appendix A with a fixed seed and is injected as a generated macro; none is typed.

2 Setting and definitions

2.1 Selective classification

Let $(x, y) \in \mathbb{R}^d \times \{1, \dots, K\}$ and let h be a classifier with confidence score $g : \mathbb{R}^d \rightarrow \mathbb{R}$ (higher means more confident). For a threshold t , the selective classifier predicts $h(x)$ when $g(x) \geq t$ and abstains otherwise. On a test sample of size n , sorting by decreasing g and accepting the first i points gives coverage $c_i = i/n$ and selective risk r_i (the error rate among the accepted points). The risk–coverage curve is $i \mapsto (c_i, r_i)$ and

$$\text{AURC}(g, h) = \frac{1}{n} \sum_{i=1}^n r_i,$$

the discrete version of the area under the curve [11]. Ties in g are frequent for vote-based scores; we compute the *expected* r_i under uniformly random tie-breaking, exactly, block by block (Appendix A). A score with no information about correctness yields, in expectation, r_i equal to the overall error rate at every coverage, so the error rate of h is the AURC of a random ordering and serves as the “no information” level. We also report the selective accuracy $1 - r_i$ at coverages 0.9 and 0.8.

2.2 The anisotropic resolution field

Definition 2.1 (Resolution field and derived scores). Let $D = \{(x_j, y_j)\}_{j=1}^N$ be a training set with standardised features. Fix a neighbourhood size K_m and a shrinkage $\alpha \in (0, 1]$. For a query x , let $N(x)$ be its K_m nearest training points in the Euclidean metric, $C(x)$ their sample covariance, $\tau(x) = \text{tr } C(x)/d$, and

$$S(x) = (1 - \alpha) C(x) + \alpha \tau(x) I, \quad A(x) = S(x)^{-1}.$$

$A(x)$ is the *resolution field*; the ellipsoid $\{u : u^\top A(x) u \leq 1\}$ is the *resolution ellipsoid* at x . For each class c , the *local prototype* $\mu_c(x)$ is the mean of the points of class c in $N(x)$, or the nearest training point of class c when the class is absent from $N(x)$. With $d_c^2(x) = (x - \mu_c(x))^\top A(x) (x - \mu_c(x))$ and $d_{(1)}^2 \leq d_{(2)}^2 \leq \dots$ the sorted values, the field classifier predicts $\arg \min_c d_c^2(x)$ and the three scores are

$$s_{\text{margin}}(x) = d_{(2)}^2(x) - d_{(1)}^2(x), \quad s_{\text{vol}}(x) = -\log \det S(x), \quad s_{\text{anis}}(x) = -\log \frac{\lambda_{\max}(S(x))}{\lambda_{\min}(S(x))}.$$

The margin is the distance to the local decision boundary between the two nearest prototypes, measured in resolution units. The log-volume is a pure density proxy (a large ellipsoid means sparse data, hence low resolution); the anisotropy score ranks elongated neighbourhoods as less certain. Two ablations isolate the geometry: the *isotropic field* sets $\alpha = 1$, so $A(x) = \tau(x)^{-1}I$ and the margin is the Euclidean margin in units of the local scale; the *Euclidean field* sets $A(x) = I$, which is nearest-local-centroid classification. A further variant keeps $A(x)$ but uses the global class means as prototypes. The shrinkage towards a scaled identity is the standard device for neighbourhood covariances in moderate dimension [18]; α and K_m are tuned.

2.3 References

All references are tuned by inner cross-validation on the same criterion (AURC) and see the same preprocessed features. (a) k -NN with vote-fraction confidence, uniform or distance weights, $k \in \{3, \dots, 31\}$. (b) Prototype methods: nearest class mean (NCM) with Euclidean margin; LDA (prototypes with the global pooled metric $\hat{\Sigma}^{-1}$) and QDA (class-specific metrics), both with maximum-posterior confidence; the LVQ family [17] is represented by NCM. (c) A locally adaptive metric: discriminant adaptive nearest neighbours (DANN) [13], one iteration, $\Sigma(x) = W^{-1/2}[W^{-1/2}BW^{-1/2} + \varepsilon I]W^{-1/2}$ from the within- and between-class scatter of $K_m \in \{50, 100\}$ neighbours, then a k -NN vote under $\Sigma(x)$. (d) Maximum softmax probability of tuned logistic regression and of a random forest [1]. Implementations are from scikit-learn [20] except DANN and the field, which are written in NumPy [12].

3 An exact statement: what anisotropy could add

The comparison is only interpretable if one knows what the field score reduces to when the metric is the “right” one. The right one, in the classical setting of Gaussian classes with a shared covariance Σ , is Σ^{-1} : the LDA rule [8, 14]. Write $d_c^2(x) = (x - \mu_c)^\top \Sigma^{-1}(x - \mu_c)$ with the true class means, $s(x) = d_{(2)}^2(x) - d_{(1)}^2(x)$ for the resolution margin under this metric, and

$$p_c(x) = \frac{\pi_c \exp(-d_c^2(x)/2)}{\sum_j \pi_j \exp(-d_j^2(x)/2)}$$

for the LDA posterior with priors π_c , whose two largest values we denote $p_{(1)}(x) \geq p_{(2)}(x)$. The *posterior margin* is $m(x) = p_{(1)}(x) - p_{(2)}(x)$.

Proposition 3.1 (Resolution margin versus LDA posterior margin). *[proved here] Let $K \geq 2$ Gaussian classes share the covariance $\Sigma \succ 0$.*

- (i) *If $K = 2$ and $\pi_1 = \pi_2$, the nearest-prototype rule $\arg \min_c d_c^2$ is the Bayes rule and $m(x) = \tanh(s(x)/4)$. Hence s and m induce the same ordering of every test set, the same risk-coverage curve and the same AURC, and the rejection rules $\{s < t\}$ and $\{m < \tanh(t/4)\}$ coincide.*
- (ii) *If $K = 2$ and $\pi_1 \neq \pi_2$, let $\sigma(x) = d_2^2(x) - d_1^2(x)$ be the signed margin. Then $\log \frac{p_1}{p_2} = \frac{\sigma}{2} + \log \frac{\pi_1}{\pi_2}$ and $m = \tanh(|\sigma + 2 \log(\pi_1/\pi_2)|/4)$. The unsigned score $s = |\sigma|$ is not a function of m , and the nearest-prototype prediction differs from the Bayes prediction on the band $\{-2 \log(\pi_1/\pi_2) < \sigma < 0\}$ (for $\pi_1 > \pi_2$); the prior-corrected score $|\sigma + 2 \log(\pi_1/\pi_2)|$ restores (i).*
- (iii) *If $K \geq 3$ and the priors are equal, $s(x) = 2 \log(p_{(1)}(x)/p_{(2)}(x))$: the resolution margin is a strictly increasing function of the log-ratio of the two largest posteriors. It is not a function of m nor of the maximum posterior $p_{(1)}$. Explicitly, with $\Sigma = I$, $\mu_1 = (0, 0)$, $\mu_2 = (2, 0)$, $\mu_3 = (1, 2)$, the points $x_A = (0.75, 0)$ and $x_B = (0.75, 0.5)$ both have $s = 1$ but $m(x_A) = 0.221$, $m(x_B) = 0.189$ and $p_{(1)}(x_A) = 0.562$, $p_{(1)}(x_B) = 0.481$.*

Proof. (i) With equal priors $\log(p_1/p_2) = (d_2^2 - d_1^2)/2$, so the Bayes rule picks the smaller d_c^2 , and $p_{(1)}/p_{(2)} = e^{s/2}$. Since $p_{(1)} + p_{(2)} = 1$, $m = (e^{s/2} - 1)/(e^{s/2} + 1) = \tanh(s/4)$, a strictly increasing bijection $[0, \infty) \rightarrow [0, 1)$. Every selective-classification quantity depends on g only through the induced ordering (including ties) and on the predictions, which agree. (ii) The first identity is the LDA log-odds; the Bayes prediction is $\text{sign}(\sigma + 2 \log(\pi_1/\pi_2))$, whereas the nearest-prototype prediction is $\text{sign } \sigma$; they differ exactly on the stated band, and two points with $\sigma = \pm a$ have the same $s = a$ but different m unless $\log(\pi_1/\pi_2) = 0$. (iii) With equal priors the ranking of the posteriors is the reverse ranking of the d_c^2 , so $p_{(1)}/p_{(2)} = \exp((d_{(2)}^2 - d_{(1)}^2)/2) = e^{s/2}$. For the counterexample, $(d_1^2, d_2^2, d_3^2)(x_A) = (0.5625, 1.562, 4.062)$ and $(d_1^2, d_2^2, d_3^2)(x_B) = (0.8125, 1.812, 2.312)$: in both cases classes 1, 2 are the two nearest with gap 1, but the third distance differs and enters the normalisation of the posteriors. The numbers were also checked by `experiments/lda_identity.py`. $\square$

Proposition 3.2 (Departure from LDA under an arbitrary field). *[proved here] Let $A(x) \succ 0$ be any field, $d_{A,c}^2(x) = (x - \mu_c)^\top A(x)(x - \mu_c)$, and suppose the two nearest prototypes at x under $A(x)$ and under Σ^{-1} are the same pair, with difference $\delta = \mu_{(1)} - \mu_{(2)}$ and midpoint $\bar{m} = (\mu_{(1)} + \mu_{(2)})/2$. Then*

$$s_A(x) - s(x) = 2(x - \bar{m})^\top (A(x) - \Sigma^{-1}) \delta.$$

In particular: (a) $s_A(x) = s(x)$ whenever $(A(x) - \Sigma^{-1})\delta = 0$, so components of the metric orthogonal to the discriminant direction δ cannot change the score; (b) for the isotropic rescaling $A(x) = \lambda(x)\Sigma^{-1}$ one gets $s_A = \lambda s$, the predictions are unchanged, and the ordering changes as soon as λ is non-constant (two points with $s(x) < s(x')$ swap order when $\lambda(x)/\lambda(x') > s(x')/s(x)$).

Proof. With $M = A(x) - \Sigma^{-1}$, $s_A - s = (x - \mu_{(2)})^\top M(x - \mu_{(2)}) - (x - \mu_{(1)})^\top M(x - \mu_{(1)}) = 2x^\top M\delta - (\mu_{(1)} - \mu_{(2)})^\top M(\mu_{(1)} + \mu_{(2)}) = 2(x - \bar{m})^\top M\delta$. (a) is immediate; (b) follows from $M = (\lambda - 1)\Sigma^{-1}$ and $2(x - \bar{m})^\top \Sigma^{-1}\delta = d_{(2)}^2 - d_{(1)}^2 = s$. $\square$

Remark 3.3 (What the experiment can and cannot show). Proposition 3.1(i) says that on shared-covariance data with two balanced classes the field cannot improve on LDA by construction: if $A(x)$ is estimated well it reproduces the LDA ordering; any difference is estimation error or the effect of local prototypes. Proposition 3.2 says that where the field *could* help, it must do so through the component of $A(x) - \hat{\Sigma}^{-1}$ along the local discriminant direction, which is exactly the quantity that class-dependent covariances (QDA regime) or curved boundaries make non-zero. This dictates the synthetic datasets: one shared-covariance control where no gain is possible, one class-dependent-covariance problem and one curved-boundary problem with anisotropic noise where a gain is possible in principle, and one generic problem with redundant features. [proved here / design]

Remark 3.4 (Heuristic: the unsupervised field is biased along δ). $C(x)$ is the covariance of the *mixture* of classes in $N(x)$. Where two classes meet it equals the within-class covariance plus a between-class term proportional to $\delta\delta^\top$, so $A(x)$ is *smaller* along δ precisely near boundaries, shrinking the margin there. The direction of this bias is consistent with abstention, but its size is set by the local class balance rather than by the class geometry; DANN separates the within- and between-class parts and avoids it. This is offered as an explanation of the results, not as a proved statement. [heuristic]

The numerical verification (`experiments/lda_identity.py`, true parameters, $d = 5$, $n = 4000$ points per case) gives: for $K = 2$ and equal priors, Spearman correlation between s and m equal to 1.000000, $\max |m - \tanh(s/4)| = 2.8 \times 10^{-16}$, AURC difference 0; for priors (0.8, 0.2), Spearman 0.9725, monotone: no, predictions differ on 0.57% of the points, and the corrected score is monotone (yes, deviation 4.4×10^{-16}); for $K = 3$ (error rate 11.2%),

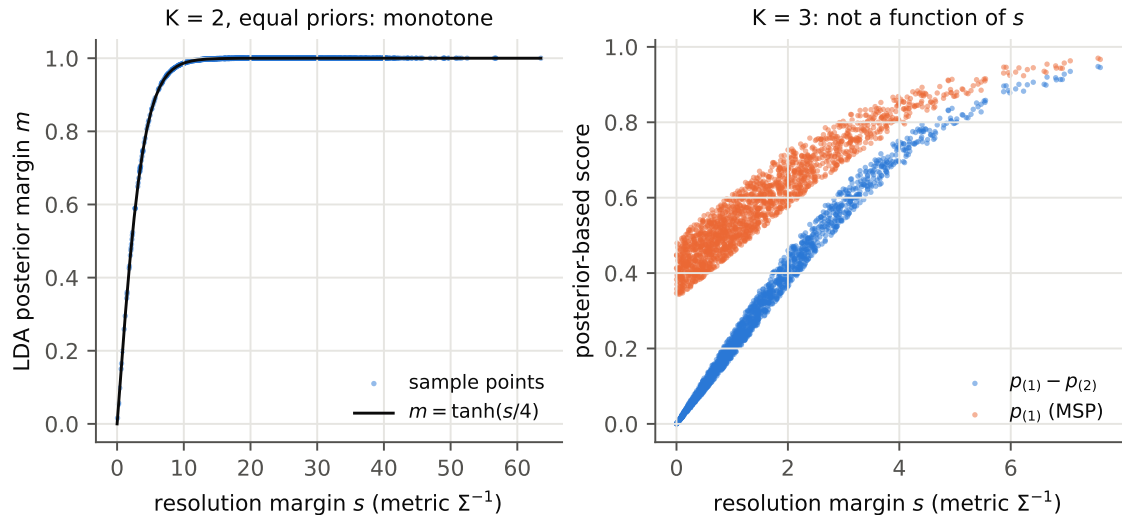


Figure 1: Proposition 3.1 on samples from the true model. Left: $K = 2$, equal priors, the posterior margin is exactly $\tanh(s/4)$. Right: $K = 3$, neither the posterior difference nor the maximum posterior is a function of s .

$\max |s - 2 \log(p_{(1)}/p_{(2)})| = 7.1 \times 10^{-15}$, monotone in the log-ratio: yes, in m : no, in $p_{(1)}$: no, with AUC $\times 100$ of 2.415 (s), 2.392 (m) and 2.380 ($p_{(1)}$): the three orderings are different but nearly equivalent in this instance (Spearman 0.9993 and 0.9971). Figure 1 shows both cases. [computationally verified]

4 Experimental protocol and predefined criterion

Datasets. Table 1. The real datasets are those shipped with scikit-learn [8, 20, 21]; digits is reduced to 20 principal components (fitted on each training fold) so that neighbourhood covariances are estimable for all local methods alike. The synthetic problems follow Remark 3.3: *synth-informative* is `make_classification` with informative, redundant and noise features; *moons-aniso* adds anisotropic Gaussian noise (standard deviations 0.30 and 0.06 along axes rotated by 30°) and three nuisance dimensions to the two-moons problem; *synth-classcov* has two Gaussian classes in $d = 6$ with the same covariance eigenvalues but different random rotations; *synth-lda* has three Gaussian classes with one shared covariance. In the last two the mean separation is chosen by bisection on a Monte-Carlo estimate of the Bayes error, with target 10% and search interval $[0.05, 20]$. For *synth-lda* this gives a shift of 1.20 and a Bayes error of 9.8%. For *synth-classcov* the two rotated covariances alone separate the classes better than the target, so the bisection stops at the lower bound (shift 0.05) and the Bayes error is 7.7%: the problem is, to within that residual shift, a pure covariance-difference problem in which the class means coincide. This was not intended but is kept, because it is the sharpest version of the regime where a local metric could matter, and it explains why the linear references are at chance level there.

Evaluation. Repeated stratified 5-fold cross-validation with 3 repeats (15 outer folds, identical for all methods). Inside each training fold, features are standardised and every method’s hyperparameters are chosen by 3-fold inner cross-validation minimising AUC; the chosen configuration is refitted on the whole training fold and scored on the test fold. Paired per-fold differences of AUC are summarised by the mean and a percentile bootstrap 95% interval over folds ($B = 2000$) [6], with a two-sided Wilcoxon signed-rank p -value as a secondary check [4]. Folds of a repeated cross-validation are not independent, so the intervals are somewhat optimistic [5, 19];

dataset	n	d	K	Bayes err. (%)	description
iris	150	4	3	–	Fisher’s iris
wine	178	13	3	–	UCI wine
breast-cancer	569	30	2	–	Wisconsin diagnostic
digits	1797	20	10	–	UCI optical digits, PCA to 20 comp.
synth-informative	1200	10	2	–	<code>make_classification</code> : 5 inf. + 3 red. + 2 noise, 2 clusters/class, 3% label noise
moons-aniso	1200	5	2	–	two moons + anisotropic noise (sd 0.30×0.06 , 30°) + 3 nuisance dims
synth-classcov	1200	6	2	7.7	two Gaussians, $d = 6$, rotated class-specific covariances
synth-lda	1200	6	3	9.8	three Gaussians, $d = 6$, one shared covariance

Table 1: Datasets. d is the dimension after preprocessing; the Bayes error is known only for the Gaussian problems.

dataset	k -NN	NCM	LDA	QDA	LogReg	RForest	DANN	err.
iris	0.68	2.73	0.17	0.23	0.25	0.54	0.19	2.0
wine	0.69	0.57	0.10	0.03	0.11	0.15	0.70	0.9
breast-cancer	0.57	1.15	0.61	0.54	0.25	0.64	0.86	2.3
digits	0.28	2.97	1.54	1.80	0.64	0.41	0.25	2.5
synth-informative	4.81	21.04	18.00	6.41	17.54	3.51	3.96	10.5
moons-aniso	0.82	2.36	2.18	2.23	2.17	0.62	1.23	4.0
synth-classcov	1.69	48.46	48.44	0.97	48.45	2.48	1.77	7.3
synth-lda	2.90	5.05	1.58	1.64	1.63	3.67	2.26	8.9

Table 2: References: $\text{AURC} \times 100$, mean over 15 folds (lower is better; best in bold). “err.” is the error rate $\times 100$ of the best reference at full coverage, i.e. the AUC of an uninformative score for that classifier.

this works *against* a negative conclusion, not for it.

Predefined criterion. Fixed before the run: the field score shows a *residual gain* if its AUC is lower than that of the best reference (the reference with the lowest mean AUC on that dataset, among the 7 references) on a majority (≥ 5 of 8) of datasets, with the paired 95% interval excluding zero. Choosing the best reference per dataset on the same folds favours the references slightly; we therefore also report the field against each reference separately. Total compute: 449 s of CPU (7.5 min) for the benchmark, single-threaded, plus 0.0 s for the verification of Section 3.

5 Results

5.1 Main comparison

Tables 2–4 and Figure 2 give the main result. Against the best reference, the anisotropic field is significantly better on 0 of 8 datasets (none), significantly worse on 6 (iris, breast-cancer, synth-informative, moons-aniso, synth-classcov, synth-lda) and inconclusive on 2 (wine, digits). The predefined criterion is therefore **not met**. The closest the field comes to the best reference is on digits ($\Delta = +0.01$); the largest loss is on synth-informative ($\Delta = +1.30$). The isotropic and Euclidean ablations meet the criterion on 0 and 0 datasets respectively, and the geometry-only scores on 0 and 0.

Table 5 removes the best-reference selection. Counting datasets on which the field is significantly better/worse: k -NN 2/1, NCM 7/0, LDA 4/3, QDA 3/3, logistic regression 4/2, random forest 3/2, DANN 2/3. The field does beat the global-prototype methods (NCM, LDA, logistic regression) where their linear boundaries are wrong (synth-classcov, synth-informative,

dataset	best ref.	AURC	aniso	iso	Eucl.	volume	anisotropy	aniso, glob. prot.
iris	LDA	0.17	0.37	0.65	0.76	1.54	6.79	1.77
wine	QDA	0.03	0.08	0.28	0.34	0.65	0.64	1.10
breast-cancer	LogReg	0.25	0.83	0.45	0.57	4.73	4.35	1.66
digits	DANN	0.25	0.26	1.04	4.11	0.66	1.42	2.16
synth-informative	RForest	3.51	4.81	4.95	4.95	17.10	15.60	21.31
moons-aniso	RForest	0.62	1.19	1.23	1.64	4.67	10.18	2.00
synth-classcov	QDA	0.97	1.40	1.24	1.23	18.02	16.69	46.62
synth-lda	LDA	1.58	2.83	2.94	3.10	15.02	14.42	4.48

Table 3: Field variants: $\text{AURC} \times 100$, mean over 15 folds, next to the best reference of Table 2. “aniso” is the full method of Definition 2.1; “iso” and “Eucl.” are its ablations; “volume” and “anisotropy” rank by geometry alone (with the predictions of the full method).

dataset	best ref.	Field (aniso)		Field (iso)		Field (Eucl.)	
		Δ [95% CI]	wins	Δ [95% CI]	wins	Δ [95% CI]	wins
iris	LDA	+0.20 [+0.04, +0.43]	1/15	+0.48 [+0.24, +0.77]	0/15	+0.59 [+0.30, +0.92]	0/15
wine	QDA	+0.05 [-0.00, +0.10]	4/15	+0.25 [+0.10, +0.42]	2/15	+0.31 [+0.11, +0.53]	2/15
breast-cancer	LogReg	+0.58 [+0.35, +0.89]	0/15	+0.21 [+0.08, +0.35]	1/15	+0.33 [+0.19, +0.47]	1/15
digits	DANN	+0.01 [-0.11, +0.13]	7/15	+0.79 [+0.34, +1.31]	2/15	+3.86 [+3.08, +4.78]	0/15
synth-informative	RForest	+1.30 [+0.81, +1.85]	2/15	+1.44 [+0.93, +1.92]	2/15	+1.44 [+0.98, +1.90]	2/15
moons-aniso	RForest	+0.56 [+0.34, +0.75]	2/15	+0.61 [+0.38, +0.83]	2/15	+1.02 [+0.76, +1.28]	0/15
synth-classcov	QDA	+0.43 [+0.32, +0.55]	0/15	+0.28 [+0.19, +0.38]	0/15	+0.26 [+0.16, +0.36]	1/15
synth-lda	LDA	+1.25 [+1.04, +1.51]	0/15	+1.36 [+1.07, +1.68]	0/15	+1.52 [+1.22, +1.85]	0/15

Table 4: Paired differences $\Delta = \text{AURC}(\text{field}) - \text{AURC}(\text{best reference})$, $\times 100$, with percentile bootstrap 95% intervals over folds, and the number of folds in which the field had the lower AURC. Negative Δ favours the field.

moons-aniso), and it beats the non-parametric local references (k -NN, DANN, random forest) on synth-classcov; but on every dataset at least one reference is at least as good, and on synth-classcov that reference is QDA, whose class-specific metric is the population version of what the field estimates locally. Moreover the gain over k -NN and DANN on synth-classcov is not attributable to anisotropy: the Euclidean ablation of the field (local prototypes, identity metric) reaches 1.23 against 1.40 for the full method (Table 3), so what helps there is the local-prototype margin, not the metric.

5.2 Ablations: does anisotropy add information?

Table 6 answers the line’s question in its narrow form. Allowing anisotropy improves on the isotropic field (same neighbourhoods, same prototypes, scalar metric) with an interval excluding zero on 3 datasets (iris, wine, digits) and hurts on 2 (breast-cancer, synth-classcov); against the Euclidean field the counts are 4 and 1. So the anisotropic metric does carry information about correctness that a local scale alone does not. Replacing the local prototypes by the global class means while keeping $A(x)$ is significantly worse on 8 datasets: the local metric is meaningful only at the scale of the neighbourhood from which it was estimated. Finally, the field is significantly worse than k -NN on 2 datasets and better on 2; worse than DANN on 3 and better on 2; worse than LDA on 3 and better on 4.

Table 7 shows that the two purely geometric scores are far from the margin. Ranking by log-volume is *worse* than a random ordering of the same predictions on 3 of 8 datasets (synth-informative, synth-classcov, synth-lda) and above half of the random level on 5; ranking by anisotropy is worse than random on 5 (iris, synth-informative, moons-aniso, synth-classcov, synth-lda). The margin is below half of the random level on every dataset. The information the

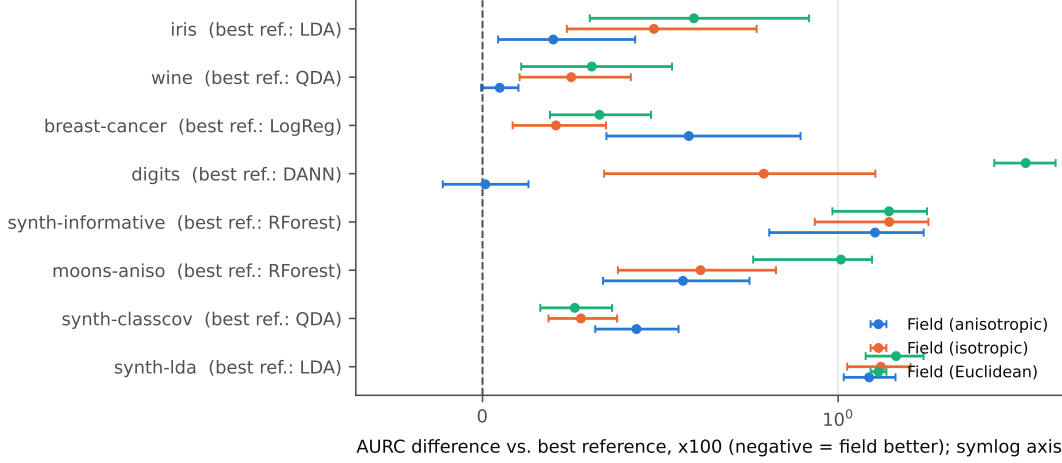


Figure 2: Paired AUC differences (field minus best reference, $\times 100$) with 95% bootstrap intervals; negative favours the field. The horizontal axis is symmetric-log to accommodate the large losses of the ablations.

dataset	vs k -NN	vs NCM	vs LDA	vs QDA	vs LogReg	vs RForest	vs DANN
iris	-0.31	-2.36	<i>+0.20</i>	<i>+0.14</i>	+0.11	-0.17	<i>+0.17</i>
wine	-0.61	-0.49	-0.02	+0.05	-0.03	-0.07	-0.61
breast-cancer	+0.25	-0.32	<i>+0.21</i>	+0.29	<i>+0.58</i>	+0.18	-0.04
digits	-0.02	-2.71	-1.28	-1.54	-0.38	-0.15	+0.01
synth-informative	+0.01	-16.23	-13.18	-1.60	-12.72	<i>+1.30</i>	<i>+0.86</i>
moons-aniso	<i>+0.37</i>	-1.18	-0.99	-1.05	-0.99	<i>+0.56</i>	-0.05
synth-classcov	-0.29	-47.06	-47.04	<i>+0.43</i>	-47.05	-1.08	-0.37
synth-lda	-0.07	-2.22	<i>+1.25</i>	<i>+1.19</i>	<i>+1.20</i>	-0.84	<i>+0.56</i>

Table 5: Anisotropic field against each reference separately: mean paired Δ of AUC $\times 100$ (negative favours the field). Bold: field significantly better (95% interval below 0); italic: significantly worse; roman: inconclusive.

field carries about correctness is therefore in the margin, i.e. in *where the prototypes are* relative to the ellipsoid, and not in the ellipsoid’s size or shape by itself. [computationally verified]

5.3 Risk–coverage curves, fixed coverage, selected hyper-parameters

Figure 3 and Table 8 show the same picture at fixed coverage. On 5 datasets (iris, breast-cancer, synth-informative, moons-aniso, synth-lda) the field’s classifier is already less accurate than the best reference at full coverage, by +1.8 (iris), +4.4 (breast-cancer), +3.0 (synth-informative), +4.9 (moons-aniso) and +2.9 (synth-lda) percentage points, so part of the AUC gap is a weaker classifier rather than a weaker ranking. On digits the field is slightly *more* accurate than DANN at full coverage (97.7% against 97.5%) yet behind at 90% coverage (99.2% against 99.7%), so there the ranking is the weaker part; the two effects cancel in the AUC. The hyper-parameters selected for the anisotropic field over all 120 dataset–fold pairs were $\alpha = 0.05$ in 9% of the cases, $\alpha = 0.2$ in 18% and $\alpha = 0.5$ in 72%; $K_m = 20$ in 31%, 40 in 16% and 80 in 53%: the tuning prefers the strongest shrinkage on offer, i.e. a mildly anisotropic field. Figure 4 illustrates the estimated ellipsoids on moons-aniso.

dataset	aniso—iso	aniso—Eucl.	Eucl.—NCM	aniso—vol.	aniso—glob. prot.	aniso— k -NN	aniso—LDA	aniso—QDA
iris	-0.28	-0.40	-1.97	-1.18	-1.40	-0.31	<i>+0.20</i>	
wine	-0.20	-0.26	-0.23	-0.56	-1.02	-0.61	-0.02	
breast-cancer	<i>+0.37</i>	+0.25	-0.57	-3.90	-0.84	<i>+0.25</i>	<i>+0.21</i>	
digits	-0.78	-3.85	<i>+1.14</i>	-0.40	-1.91	-0.02	-1.28	
synth-informative	-0.14	-0.14	-16.09	-12.28	-16.49	+0.01	-13.18	
moons-aniso	-0.05	-0.46	-0.72	-3.49	-0.81	<i>+0.37</i>	-0.99	
synth-classcov	<i>+0.16</i>	<i>+0.17</i>	-47.24	-16.62	-45.22	-0.29	-47.04	
synth-lda	-0.11	-0.27	-1.95	-12.19	-1.66	-0.07	<i>+1.25</i>	

Table 6: Ablations and direct comparisons: mean paired difference of $\text{AURC} \times 100$ (first minus second; negative favours the first). Bold: first significantly better; italic: second significantly better.

dataset	margin		volume		anisotropy	
	err.	AURC	err.	AURC	err.	AURC
iris	3.8	0.37	4.2	1.54	4.9	<i>6.79</i>
wine	1.7	0.08	2.6	0.65	2.2	0.64
breast-cancer	6.7	0.83	6.6	4.73	6.6	4.35
digits	2.3	0.26	2.2	0.66	2.1	1.42
synth-informative	13.6	4.81	14.0	<i>17.10</i>	14.1	<i>15.60</i>
moons-aniso	9.0	1.19	9.1	4.67	8.9	<i>10.18</i>
synth-classcov	8.3	1.40	9.1	<i>18.02</i>	9.1	<i>16.69</i>
synth-lda	11.8	2.83	13.0	<i>15.02</i>	12.3	<i>14.42</i>

Table 7: Geometry-only scores ($\times 100$). For each variant, “err.” is the error rate of its own (tuned) field classifier at full coverage, i.e. the AURC of a random ordering of its predictions; italics mark an AURC at or above that level. Volume and anisotropy rank the predictions by the shape of $S(x)$ alone.

6 What can be claimed

Table 9 lists the claims with their status. The main one is negative and is stated as such: under a criterion fixed in advance, on 8 datasets chosen to include the regimes where a local anisotropic metric could help, the resolution-field score does not deliver a residual gain over tuned local-metric, prototype and softmax references. The one regime in which the field is competitive, class-dependent covariance, is exactly the regime in which a parametric reference (QDA) is available and better. The positive finding is narrow: the anisotropic metric ranks correctness better than an isotropic local scale on 3 datasets, so the representation is not empty; it is redundant given the references.

Limitations. The field is one specific reconstruction of an idea for which no manuscript existed; other formulations (a supervised field from the Fisher information of a local probabilistic model, a field learned end-to-end, or a combination of the margin with the log-volume) were not run, and the second and third could in principle change the outcome. The datasets are small to moderate ($n \leq 1797$, $d \leq 30$) and low-dimensional after preprocessing; neighbourhood covariances in higher dimension would need stronger regularisation and the comparison could shift either way. Inner tuning on AURC with 3 folds is noisy on the smallest datasets. Bootstrap intervals over repeated folds are optimistic, and the best reference is selected on the same folds; both caveats make the negative result conservative in the sense that they favour finding a gain. DANN was run for one iteration with $\varepsilon = 1$ as in the original paper and not tuned further. QDA at $\text{reg_param} = 0$ was excluded because it fails on rank-deficient inner folds. The synth-classcov problem ended with coinciding class means (Section 4); a version with both a mean shift and different covariances was not run.

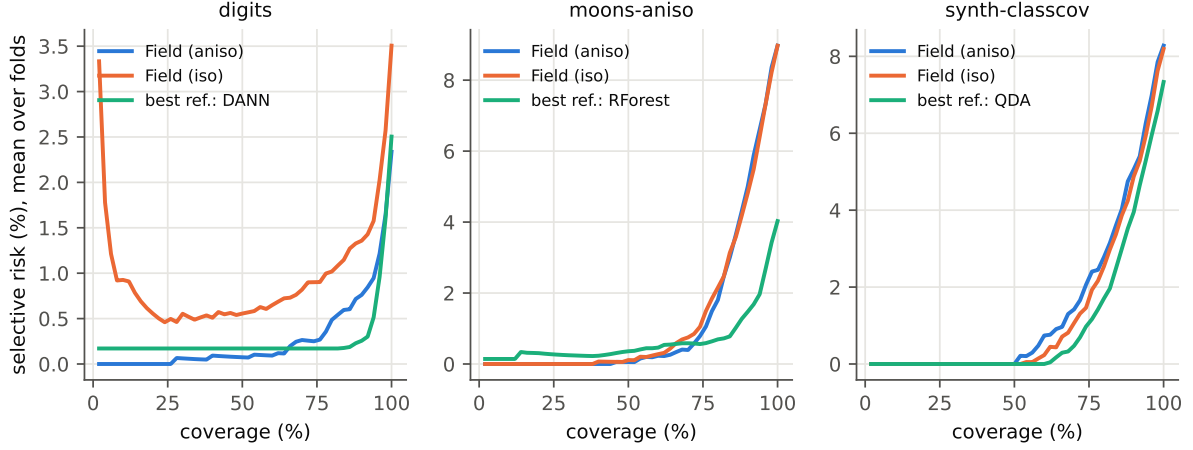


Figure 3: Mean risk-coverage curves over folds on three datasets, for the anisotropic field, its isotropic ablation and the best reference.

dataset	best ref.	best ref.			k -NN			Field (aniso)			Field (iso)		
		100	90	80	100	90	80	100	90	80	100	90	80
iris	LDA	98.0	99.8	100.0	96.2	98.0	99.2	96.2	98.8	99.7	95.8	97.8	98.9
wine	QDA	99.1	100.0	100.0	96.6	98.6	99.5	98.3	100.0	100.0	96.4	99.2	99.8
breast-cancer	LogReg	97.7	99.2	99.7	96.0	98.6	99.5	93.3	97.2	98.8	96.0	98.3	99.5
digits	DANN	97.5	99.7	99.8	96.8	99.3	99.8	97.7	99.2	99.5	96.5	98.6	99.0
synth-informative	RForest	89.5	92.4	94.7	86.7	90.0	92.7	86.4	89.6	92.6	85.9	89.1	92.1
moons-aniso	RForest	96.0	98.5	99.3	94.3	97.5	98.9	91.0	95.0	98.2	91.0	95.2	97.8
synth-classcov	QDA	92.7	96.0	98.3	89.8	94.1	96.7	91.7	94.9	97.2	91.8	95.1	97.5
synth-lda	LDA	91.1	94.8	97.5	88.8	92.9	95.5	88.2	91.8	95.0	88.1	92.1	94.7

Table 8: Accuracy (%) at coverage 100%, 90% and 80%, mean over folds.

Next steps for the research line. If the line is reopened at all, the only directions consistent with the results above are: (a) a supervised field, $A(x)$ built from within-class rather than mixture covariances (the DANN decomposition) with the field margin rather than a vote as the score, tested on synth-classcov and moons-aniso first, since these are the regimes where Proposition 3.2 allows a gain; (b) the margin combined with the log-volume as a two-dimensional rejection rule, evaluated under the same criterion; (c) a higher-dimensional benchmark with explicit covariance regularisation. Absent these, the line is closed: the field is a valid but redundant representation of uncertainty for abstention on the problems studied.

A Reproducibility

The directory contains `experiments/selective_benchmark.py` (benchmark; `-fast` for a reduced run), `experiments/lda_identity.py` (verification of Section 3), `experiments/make_figures.py` and `experiments/make_numbers.py`, which converts `results/results.json` and `results/results_identity.json` into the macros and table bodies of this document. Seed 20260930; Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1 [23], scikit-learn 1.9.1, Matplotlib [16]; BLAS threads limited to one; 449 s of CPU for the benchmark (wall 477 s).

Tie handling. Sort by decreasing score, group tied scores into blocks; within a block of size b with e errors, the expected cumulative error after accepting $j \leq b$ of its points is ej/b , and r_i is the expected cumulative error divided by i . This gives the exact expectation of the risk-coverage

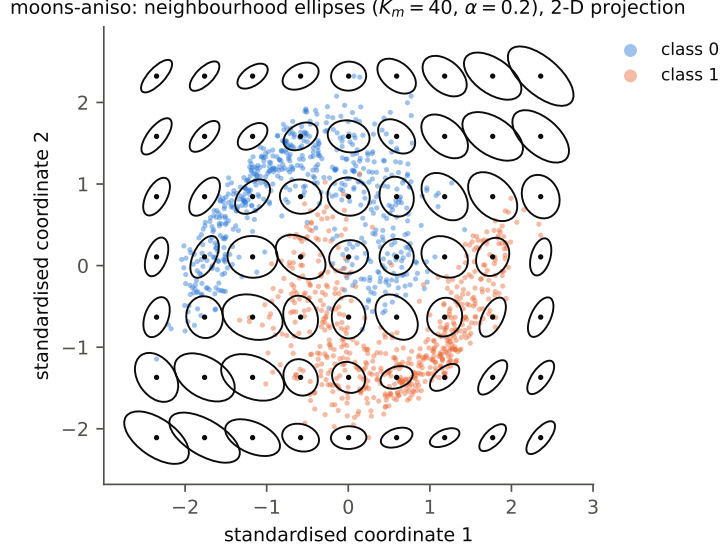


Figure 4: Resolution ellipsoids ($K_m = 40$, $\alpha = 0.2$, drawn at 0.6 standard deviations, projected on the two informative coordinates) on a grid of query points of moons-aniso. Along the moons the ellipses follow the local orientation of the data; away from them they inflate.

curve under uniform random tie-breaking; the AURC is its mean over i and the accuracy at coverage c is $1 - r_{[cn]}$.

Grids. k -NN: $k \in \{3, 5, 7, 11, 15, 21, 31\}$, uniform or inverse-distance weights. LDA: least-squares solver, shrinkage none or Ledoit–Wolf. QDA: $\text{reg_param} \in \{0.01, 0.1, 0.5\}$. Logistic regression: $C \in \{0.1, 1, 10\}$, L-BFGS, 3000 iterations. Random forest: 100 trees, $\sqrt{d}$ features per split, minimum leaf size $\{1, 5\}$. DANN: $K_m \in \{50, 100\}$ (capped at $N - 1$), $k \in \{5, 11, 21\}$, $\varepsilon = 1$, ridge $10^{-3} \text{tr } W/d$ on W . Field: $K_m \in \{20, 40, 80\}$, $\alpha \in \{0.05, 0.2, 0.5\}$ (isotropic ablation $\alpha = 1$; Euclidean ablation $A = I$). Inner selection minimises the mean inner AURC; the hyper-parameter key carries the nominal K_m so that inner and outer configurations match on small folds.

Datasets. Standardisation is fitted on each training fold; digits is projected on 20 principal components fitted on the training fold. Synthetic generators are in `make_datasets`; the class-mean separation of synth-classcov and synth-lda is found by 40 bisection steps on a Monte-Carlo estimate (40 000 points) of the Bayes error.

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claim	status
LDA posterior and Bayes optimality for shared-covariance Gaussians	classical
$K = 2$, equal priors: resolution margin under Σ^{-1} is $\tanh^{-1}$ -related to the posterior margin; identical risk–coverage curves (Prop. 3.1(i))	proved here
Unequal priors: identity fails for the unsigned margin; restored by the prior-corrected signed margin (Prop. 3.1(ii))	proved here
$K \geq 3$: margin equals twice the top-two posterior log-ratio, not a function of the posterior difference or of the maximum posterior (Prop. 3.1(iii))	proved here, example verified
Departure from LDA under any field is $2(x - \bar{m})^\top (A - \Sigma^{-1})\delta$; only the component along δ matters (Prop. 3.2)	proved here
Unsupervised local covariance is inflated along δ near boundaries	heuristic
Predefined criterion for a residual gain of the field over the best reference: not met (0 better, 6 worse, 2 inconclusive of 8)	computationally verified
Anisotropy improves on the isotropic ablation on 3 datasets and hurts on 2	computationally verified
The field beats k -NN, DANN and the linear references on synth-classcov but not QDA, and its Euclidean ablation does as well: the gain is from local prototypes, not anisotropy	computationally verified, regime-specific
Geometry-only scores (volume, anisotropy) are close to uninformative	computationally verified
Behaviour on other data, in high dimension, or with supervised field estimation	not addressed
Any statement about physical or quantum phenomena	not made

Table 9: Claims and their status. “Computationally verified” means reproduced by the scripts of Appendix A with the fixed seed on the listed datasets; it is not a proof and does not generalise beyond them.

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